Python Notes

Beginner-friendly programming notes

1. Python Fundamentals

Python is a high-level, interpreted, general-purpose language known for readable syntax. It is used in automation, web development, data analysis, artificial intelligence, and machine learning.

Variables and data types

```
name = "Navaneeth"
age = 20
marks = 87.5
is_student = True
print(name, age, marks)
```

- int stores whole numbers; float stores decimal values; str stores text; bool stores True or False.
- list is ordered and mutable; tuple is ordered and immutable; set stores unique values; dict stores key-value pairs.
- Use type(value) to inspect a value's type and int(), float(), or str() for conversion.

Input, output, and operators

```
name = input("Enter name: ")
n = int(input("Enter a number: "))
print("Hello", name)
print("Square =", n ** 2)
```

- Arithmetic: +, -, *, /, //, %, **.
- Comparison: ==, !=, >, <, >=, <=.
- Logical: and, or, not.
- input() returns a string; convert it before numeric calculations.

2. Conditions, Loops, and Functions

Conditions

```
marks = 75
if marks >= 90:
print("A grade")
elif marks >= 70:
print("B grade")
else:
print("Keep practising")
```

Loops

```
for i in range(1, 6):
print(i)

n = 3
while n > 0:
print(n)
n -= 1
```

range(start, stop, step) excludes the stop value. Use break to exit a loop and continue to skip to the next iteration.

Functions

```
def add(a, b):
return a + b

print(add(4, 6))
```

Functions package reusable logic. Parameters receive input and return sends a result to the caller. Use meaningful names and keep functions focused.

3. Strings, Collections, and Files

```
word = "Python"
print(word[0])
print(word[1:4])
print(word.lower())

nums = [10, 20, 30]
nums.append(40)
student = {"name": "Ravi", "marks": 90}
print(student["name"])
```

Strings are immutable. Common string methods include lower(), upper(), strip(), replace(), split(), and find(). Lists support append(), remove(), sort(), and indexing.

File handling

```
with open("notes.txt", "w", encoding="utf-8") as f:
f.write("Learning Python")

with open("notes.txt", "r", encoding="utf-8") as f:
print(f.read())
```

The with statement closes a file automatically. Common modes are r (read), w (write or overwrite), and a (append).

Exceptions and OOP

```
try:
value = int(input("Number: "))
print(100 / value)
except ValueError:
print("Enter a valid integer")
except ZeroDivisionError:
print("Cannot divide by zero")
```

A class is a blueprint and an object is an instance. Encapsulation groups data and methods; inheritance reuses a parent class; polymorphism allows different implementations of a common interface; abstraction hides implementation details.

4. Practice Questions

1. Check whether a number is even or odd.
2. Find the largest of three numbers.
3. Calculate factorial using a loop.
4. Count vowels in a string.
5. Remove duplicates from a list.
6. Write a function that returns the largest list element.
7. Read a CSV file using pandas and show its first five rows.